INCH-POUND
MIL-DTL-12109D
20 August 1998
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31 October 1968

DETAIL SPECIFICATION

SOLENOID, ELECTRICAL; GUN-FIRING MECHANISMS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solenoids for use in gun-firing mechanisms used by the Military Services (see 6.1).

1.2 Classification. Solenoids furnished are of the following classes, as specified on the applicable specification sheet (see 6.2).

Class 1 - Solenoids for cannons.
Class 2 - Solenoids for machine guns.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5945
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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

(See Supplement 1 for list of specification sheets.)

STANDARDS

DEPARTMENT OF DEFENSE

MS33800 - Connector Receptacle, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G. Waterproof.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM G21 - Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.3 Design, materials, and manufacturing processes. Unless otherwise specified herein or on the applicable specification sheets, the design, materials and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the Government fully meet the requirements specified and the solenoids are internally grounded.

3.3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Springs. Springs shall not be used to transmit current.

3.3.3 Plunger springs. Plunger return springs shall be provided on class 2 solenoids

3.3.4 Terminals. Solenoid terminals shall conform to MS33800.

3.4 **Operating requirements.** Each solenoid shall provide the following functional, operational, and performance capabilities.

3.4.1 Voltage. The solenoids shall operate over a voltage range of 18 to 30 volts direct current (Vdc).

3.4.2 Current. The current shall not exceed the values specified on the associated specification sheets.

3.4.3 Static force. The static force shall be in accordance with the associated specification sheets.

3.4.4 Firing (class 2 only). The solenoid shall be capable of repeatedly firing the intended gun while cycling the solenoid 4 seconds (sec) on and 8 sec off for a total of 1000 rounds of ammunition without degradation of performance.

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3.4.5 Terminal strength. The solenoid cable shall be capable of withstanding a pull of 45 pounds (lbs) (20 kilograms (kg)) without permanent deformation, insulation rupture or degradation of performance.

3.4.6 Continuous duty (class 2 only). The solenoid shall maintain the applicable static force without failure during continuous operation.

3.5 **Interface requirements**. Each solenoid shall accommodate the following interface requirements.

3.5.1 Interchangeability. External dimensions shall fall within the envelope defined by of the applicable specification sheets. Electrical connectors, shaft and mounting dimensions and locations shall conform to applicable specification sheets.

3.6 **Support and ownership requirements**. Each solenoid shall meet the following requirements.

3.6.1 Safety. The solenoid shall be physically and electrically safe to operate. Clearance between any current-carrying part and other conducting parts of different potential shall be no less than 0.063 inch (in.) (1.6 millimeters (mm)) through air or insulation.

3.6.2 Identification marking. The solenoid shall prominently display at least the manufacturer's name or trademark, manufacturer's part number, date of manufacture, and PIN (military part identification number) on the outer surface, in permanent, easily visible, straight simple letters and numbers.

3.6.3 Endurance. Each solenoid shall operate for 10 000 cycles, with cycles of 0.06 sec on and 10 sec off for class 1 and 4 sec on and 8sec off for class 2 at the applicable static force without failure.

3.7 **Operating environment requirements**. Each solenoid shall operate under the following environmental conditions.

3.7.1 Waterproofness. The solenoid shall be capable of operating, submerged in saline solution, (see 4.2.2) while subjected to a 6 pounds per square inch (psi) (41 kilopascals (kPa)) differential.

3.7.2 Corrosion resistance. The solenoid shall be capable of withstanding 5% salt solution spray for at least 200 hours.

3.7.3 Fungus. The solenoid shall be capable of withstanding a fungus environment at 93 to 100% relative humidity.

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3.7.4 Vibration. The solenoid shall be capable of withstanding the severe field service vibration profile experienced in class 1 cannons and class 2 machine guns.

3.7.5 Shock. The solenoid shall be capable of withstanding a saw tooth wave shock pulse of 20 gravity units (g).

3.7.6 Sand and dust resistance. The solenoid shall be capable of withstanding a sand and/or dust laden atmosphere for at least 6 hours.

3.7.7 Temperature extremes. The solenoid shall be capable of operation in temperatures ranging from -65 to +125 degrees Fahrenheit (°F) (-53.9 to +51.7 degrees Celsius (°C)).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. When specified, first article inspection shall be performed on 8 production-representative units. First article inspection shall consists of the examinations and tests as specified in table I.

4.1.1.1 First article test sequence. First article tests shall be conducted on the samples in accordance with the test sequence specified in table II.

4.1.2 Conformance inspection (CI). Conformance inspection shall include the examinations of 4.1.2.2 and the tests of 4.1.2.3 as specified in table I. Noncompliance with any of the specified requirements shall be cause for rejection of the sample and the inspection lot.

4.1.2.1 Sampling plan. The sampling plan shall be as specified in the contract order (see 6.2).

4.1.2.2 Examination. The sample selected in accordance with 4.1.2.1 shall be physically examined as specified in table I. The major defect is interchangeability and the minor defects are identification marking and safety. The acceptance number in all cases is zero.

4.1.2.3 Tests. The sample selected in accordance with 4.1.2.1 shall be tested in accordance with table I. The current test shall be done first followed by the voltage test and continuous duty test (class 2 solenoids only).

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4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.2.1 and 4.2.2.

Table I. Verification methods.

Title	Requirement	Verification	First article	CI	
				Exam	Tests
Design, materials, and manufacturing processes	3.3 through 3.3.4	4.3.2	X		
Operating requirements	3.4	4.3.3			
Voltage	3.4.1	4.3.3.1	X		X
Current	3.4.2	4.3.3.2	X		X
Static force	3.4.3	4.3.3.3	X		
Firing (class 2 only)	3.4.4	4.3.3.4	X		
Terminal strength	3.4.5	4.3.3.5	X		
Continuous duty (class 2 only)	3.4.6	4.3.3.6			X
Interface requirements	3.5	4.3.4			
Interchangeability	3.5.1	4.3.4.1	X	X	
Support and ownership requirements	3.6	4.3.5			
Safety	3.6.1	4.3.5.1	X	X	
Identification marking	3.6.2	4.3.5.2	X	X	
Endurance	3.6.3	4.3.5.3	X		
Operating environment requirements	3.7	4.3.6			
Waterproofness	3.7.1	4.3.6.1	X		
Corrosion resistance	3.7.2	4.3.6.2	X		
Fungus	3.7.3	4.3.6.3	X		
Vibration	3.7.4	4.3.6.4	X		
Shock	3.7.5	4.3.6.5	X		
Sand and dust resistance	3.7.6	4.3.6.6	X		
Temperature extremes	3.7.7	4.3.6.7 and 4.3.6.8	X		

4.2.1 Temperature. Unless otherwise specified, the ambient air (room) temperature shall be 77 ± 15°F (25 ± 8.3°C). Specimens shall be temperature stabilized at the applicable ambient temperature for one hour before testing.

4.2.2 Saline solution. The salt used shall be sodium chloride containing on a dry basis not more than 0.1 percent (%) of sodium iodine and not more than 0.5% of the total impurities. The solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of distilled water or other water containing not more than 200 parts per million of total solids. The solution shall be kept free of sediment by filtration or decantation.

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TABLE II. Order of first article testing.

Sample no.	Verification	Paragraph
1 and 2	Terminal strength	4.3.3.5
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
	Vibration	4.3.6.4
	Voltage	4.3.3.1
	Shock	4.3.6.5
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
	Temperature extremes	4.3.6.7, 4.3.6.8
3 and 4	Terminal strength	4.3.3.5
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
	Sand and dust resistance	4.3.6.6
	Voltage	4.3.3.1
	Endurance	4.3.5.3
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
	Temperature extremes	4.3.6.7, 4.3.6.8
5 and 6	Terminal strength	4.3.3.5
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
	Fungus	4.3.6.3
	Waterproofness	4.3.6.1
	Corrosion resistance	4.3.6.2
	Current	4.3.3.2
	Voltage	4.3.3.1
	Static force	4.3.3.3
7 and 8	Terminal strength	4.3.3.5
	Firing (class 2 only)	4.3.3.4

4.3 Verification methods. The types of verification methods included in this section are visual, inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously qualified designs.

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4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.3.2 Materials, design and construction. Conformance to 3.3 through 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records may include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.3.3 **Operating requirements verification**.

4.3.3.1 Voltage. Mount the solenoid on a test stand and test with no load at 18 and 30 Vdc. The solenoid shall be cycled as specified in 4.3.3.6 for 45 minutes at each voltage. The solenoid shall operate at both 18 and 30 Vdc throughout the test without degradation in performance.

4.3.3.2 Current. Before the voltage test, verify that the current is as specified for the appropriate solenoid, using one or more of the methods outlined in 4.3.

4.3.3.3 Static force. Before and immediately after the voltage test, verify that the static force is as specified for the solenoid under test, using one or more of the methods outlined in 4.3.

4.3.3.4 Firing (class 2 only). Install the solenoid on the gun for which it is intended and fire 1000 rounds of ammunition by cycling the solenoid 4 seconds (sec) on and 8 sec off. Applied voltage shall be 18 Vdc. The solenoid shall continuously fire the gun throughout the test without failure.

4.3.3.5 Terminal strength. Exert a force of 45 lb between the solenoid and the terminal along their mutual centerline for one minute. Also, subject the cable to a pull of 32 lb (14.5 kg) along, at 45 degrees (°), and perpendicular to the centerline of the solenoid for one minute (see figure 1). The cable shall show no sign of permanent deformation exceeding 0.063 in., insulation rupture or any degradation in performance.

4.3.3.6 Continuous duty (class 2 only). The solenoid shall be mounted as in intended operation and continuously energized for 90 sec with continuous static force. Applied voltage shall be 30 Vdc and ambient air temperature 125 ± 5°F (51.7 ± 2.8°C).

4.3.4 **Interface requirements verification**.

4.3.4.1 Interchangeability. Measure the external dimensions and demonstrate placement and matching of interface (cable, mounting holes, and shafts) between the solenoid and the applicable machine guns or cannons.

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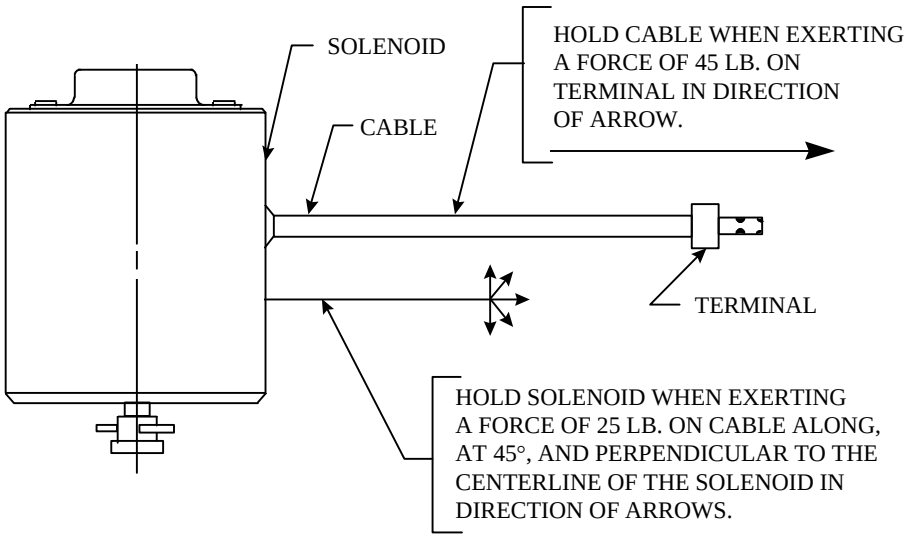


FIGURE 1. Terminal strength test.

4.3.5 **Support and ownership requirements verification**.

4.3.5.1 Safety inspection. Inspect solenoid for hazardous burrs, nicks, sharp edges, foreign materials, or other imperfections that pose physical danger to an installer/operator. Inspect solenoid for exposed, frayed, unsecured, or otherwise improperly protected circuits that pose electrical danger to an installer/operator.

4.3.5.2 Identification marking. Verify the presence of the required markings on the solenoid. After performing all environmental tests in 4.3.6, re-inspect the solenoid markings for readability.

4.3.5.3 Endurance.

a. Class 1. The solenoid shall be operated for a total of 10 000 cycles, each cycle consisting of 0.60 sec on and 10 sec off, while operating against the applicable static load, without failure. Applied voltage shall be 28 Vdc.
b. Class 2. Test the same as for class 1, except the cycle shall be 4 sec on and 8 sec off.

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4.3.6 **Operating environment requirements verification**.

4.3.6.1 Waterproofness. The solenoid, with its electrical connections, shall be submerged in a container with the uppermost surface a minimum of 1 in. below the surface of the saline solution (see 4.3.2.2) and installed in the chamber. The component shall be carefully observed during its entire period of submersion and shall be operated while submerged for 30 minutes at full rated current and voltage. The chamber shall be evacuated to a pressure 6 psi (41 kPa) below atmospheric so as to apply a minimum of 6 psi in internal pressure to all voids within the solenoid. Any evidence of bubbles escaping from the interior of the solenoid shall constitute failure of this test. Bubbles which are the result of trapped air on the exterior surfaces of the solenoid shall not be cause for failure. The chamber shall then be pressurized to 6 psi above atmospheric and the solenoid again operated for 30 minutes. The solenoid shall then be subjected to and pass the tests specified in 4.3.6.2, 4.3.3.2, 4.3.3.1 and 4.3.3.3.

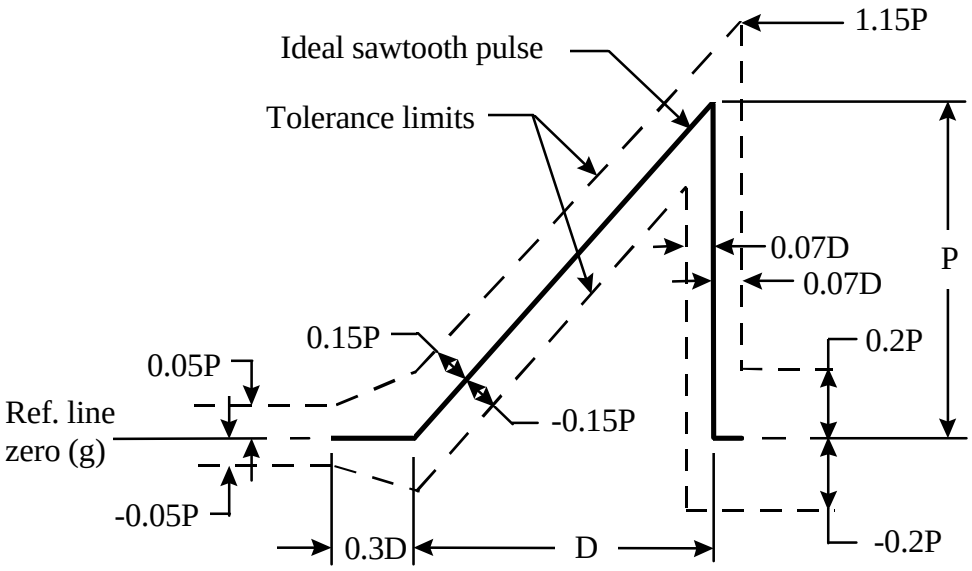
4.3.6.2 Corrosion resistance. Using spray apparatus, piping, and collector reservoir materials non-reactive to the salt spray, subject a solenoid to a continuously atomized, finely divided, wet, dense salt spray mixture of 5% sodium chloride and 95% water. Uniformly distribute the salt fog over the test article at a fall out rate between 0.01 and 0.1 ounces per 12 square inches per hour (0.3 and 3 milliliters per 77 square centimeters per hour) for at least 200 hours at a constant 95°F (35°C) with minimal air circulation. Do not use immersion heaters within the exposure area. After completing this procedure, dry the test article at ambient conditions for at least 48 hours. The solenoid shall then be subjected to and pass the voltage test specified in 4.3.3.1.

4.3.6.3 Fungus. Perform the ASTM G21 fungus resistance test, except that the test duration shall be 90 days. The solenoid shall then be subjected to and pass the voltage test specified in 4.3.3.1.

4.3.6.4 Vibration. The solenoid shall be mounted on a suitable test fixture to ensure that the mounting is free from resonance over the test frequency range. Subject the test article to simple harmonic motion having an amplitude of either 0.06 in. (0.15 cm) double amplitude (max total excursion) or 10 gravity units (g peak), whichever is less. Vary the frequency between the approximate limits of 10 to 500 Hertz (Hz). Take approximately 15 minutes to traverse the entire frequency range from 10 to 500 Hz and return to 10 Hz. Apply this motion 12 times in each of the 3 mutually perpendicular directions for a total test time of 9 hours. The solenoid shall then be subjected to and pass the voltage test specified in 4.3.3.1.

4.3.6.5 Shock. Three shocks shall be applied along the three mutually perpendicular axes of the solenoid. One of the shock axes shall be through or parallel to the solenoid plunger. The shock pulse shape shall be in accordance with figure 2 and shall have a peak value (P) of 20 g’s and a nominal duration of 10 milliseconds. The solenoid shall then be subjected to and pass the voltage test specified in 4.3.3.1.

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NOTES: D = Nominal duration in milliseconds.
P = Specified peak acceleration in g's.
g = Acceleration of gravity on earth 386 in/sec/sec (9.8 meters/sec/sec).

FIGURE 2. Sawtooth shock pulse configuration.

4.3.6.6 Sand and dust resistance. The solenoid shall be placed in test chamber with conditions maintained at 73°F (23°C) and a relative humidity of less than 22%. The solenoid shall then be subjected to a blowing dust concentration (97 to 99% by weight silicon dioxide (SiO_2)), with a particle sizes ranging from 100 to 325 micrometers, at an air velocity of 2300 ± 500 feet per minute (ft/min) (533 ± 76 meters per minute (m/min)) for at least 6 hours. After stopping the dust feeder and reducing the air velocity to 300 ± 200 ft/min (91 ± 61 m/min), the temperature shall be raised to no less than 145°F (63°C) and the humidity adjusted to maintain a relative humidity of less than 10%. These conditions shall be maintained for no less than 16 hours. The dust feeder shall then be restarted with an air velocity of 2300 ± 500 ft/min, and these conditions maintained for at least 6 hours. The test shall then be stopped and the solenoid allowed to return to standard ambient conditions. The accumulated dust shall then be removed from the test item by brushing, wiping or shaking, but care is to be taken avoid introduction of additional dust into the test item. Do not remove dust by either air blast or vacuuming. The solenoid shall then be subjected to and pass the voltage test specified in 4.3.3.1.

4.3.6.7 Low temperature test. The solenoid shall be mounted on a base and placed in a temperature test chamber at -65 ± 5°F (-53.9 ± 2.8°C) for 24 hours with the electrical circuits connected as in intended use. After 24 hours and while at this temperature, the solenoid shall be subjected to and pass the static force test specified in 4.3.3.3.

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4.3.6.8 High temperature test. Allow the solenoid to return to a temperature of 77 ± 15°F and again test as specified in 4.3.6.7, except that the ambient air temperature shall be 125 ± 5°F.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Solenoids covered by this specification are intended for use in the operation of gun-firing mechanisms but may be used for other purposes where the characteristics herein specified are applicable. These solenoids are military unique because they must be able to operate satisfactorily at ambient temperatures of -65 to 125°F. Commercial solenoids are not designed to withstand such extreme temperature conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of solenoid required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
d. When first article inspection is required (see 3.2).
e. Packaging requirements (see 5.1).

6.3 Cross reference. The following is a list of the old and new part numbers.

Old part number	New part number
M12109/5-1	M12109/5
M12109/8-1	M12109/8
M12109/13-1	M12109/13
M12109/21-1	M12109/21-1
M12109/21-2	M12109/21-2

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6.4 Subject term (key word) listing.

Continuous duty
Firing
Linear actuator
Static force

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - OS
Air Force - 85

Preparing Activity:
Army - AT

(Project 5945-1025)

Review Activities:
Army - AR, EA
Navy - MC
Air Force - 99
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-DTL-12109D	980820

3. DOCUMENT TITLE

Solenoid, Electrical; Gun-Firing Mechanisms, General Specification For

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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